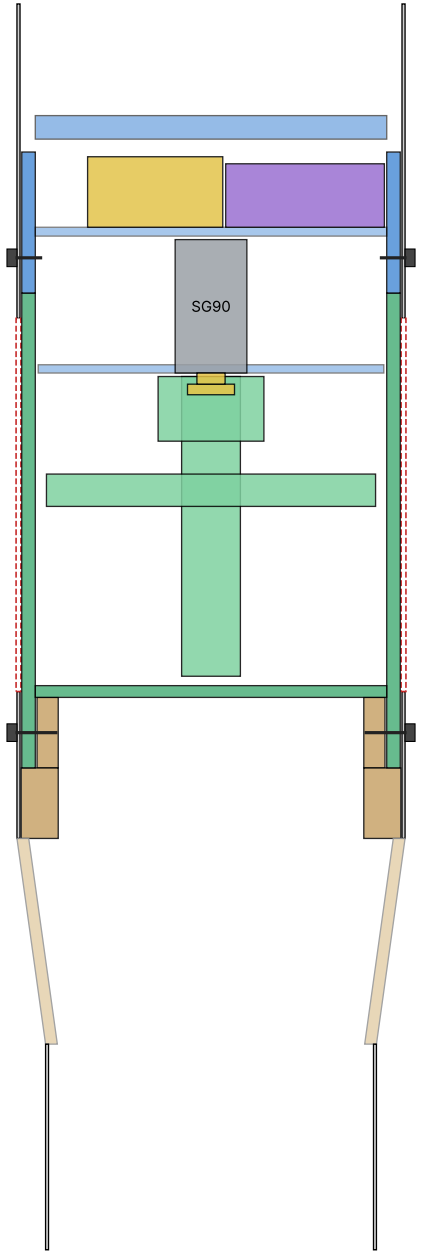
V5 sketch 2 (concept, not CAD): module bolts to the BT80 transition from the outside; BT70 side untouched

Assembled (concept): every bolt is radial, from the outside, BT80 side only



upper BT80 continues

HEAD: servo, Mercury, battery
4 radial M3 + nut

horn in the hub slot

open windows 38.6 × 63.5

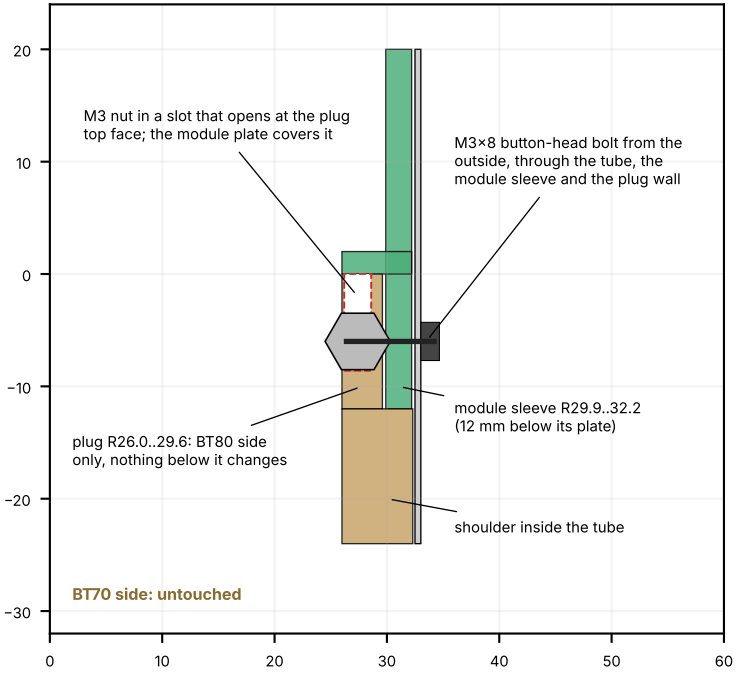
MODULE

4 radial M3 + nut: tube, module sleeve and transition plug

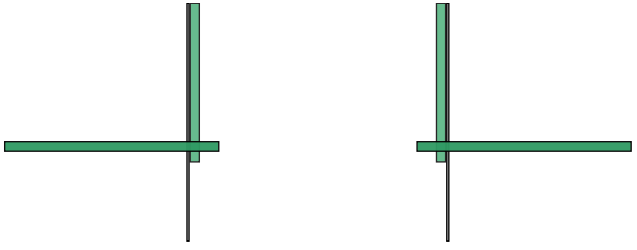
TRANSITION BT80 to BT70

BT70 side: nothing changes

Bolt joint, radial section (concept)



B open: fins out through the open windows



Screws, before and after

fastener	V4	V5
seam screws (housing halves)	4	0 (tube clamps)
catch-cap screws (rotor)	4	0 (integral caps)
hinge screws (fins)	4 M4	4 M3+nut
bottom boss screws	4	0
tube support screws	8	0
radial bolts: tube + module + transition	0	4 M3+nut
radial bolts: head to tube	0	4 M3+nut
servo M2 (SG90 parts)	3	3 M2
total	**27**	**15**

Seam without screws: the tube clamps the halves; snap hooks or a band hold them while the module slides in. Total 15 = 12 M3 + nut and 3 M2. Print-in-place hinge pins would remove 4 more (11), but the hinge gets weaker and harder to repair.

All nuts sit in slots open to the inside of the module or the head, filled while the parts are still outside the tube.

Swap on the ground, after the sections have separated:
1 pull the 4 bolts (tube + module + transition)
2 slide the transition and module out of the tube bottom
3 lift the module off the plug, set the new one on it
4 slide back in, 4 bolts. BT70 side never touched.